

Establishing coverage criteria specific to Constrained Soft Actor-Critic Algorithms

October 2026

Constrained Markov Decision Process (CMDP)

$$\max_{\pi} J_R(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t (r_t + \alpha \mathcal{H}(\pi(\cdot | s_t))) \right]$$

subject to

$$J_C(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t c_t \right] \leq d$$

- r_t : reward
- $c_t = C(s_t, a_t)$: constraint cost
- d : allowed cost budget
- γ : discount factor
- α : temperature

Environment interaction

$$(s_t, a_t) \longrightarrow (s_{t+1}, r_t, c_t) \\ a_t \sim \pi(\cdot | s_t)$$

The policy must simultaneously maximize the total return, while satisfying the cost constraint:

Lagrangian formulation

$$J(\pi, \lambda) = J_R(\pi) - \lambda(J_C(\pi) - d)$$

$$\lambda \geq 0$$

λ controls the trade-off between reward and constraint satisfaction.

What makes CSAC different?

Unlike a conventional DRL policy, CSAC simultaneously balances three competing objectives:

Reward maximization

$$Q_R(s, a)$$

The policy seeks actions with high expected reward.

Entropy maximization

$$\mathcal{H}(\pi(\cdot|s))$$

The policy maintains stochasticity and exploration.

Constraint satisfaction

$$Q_C(s, a)$$

The policy must avoid excessive cumulative cost.

The two adaptive parameters determine the relative importance of these mechanisms:

$$\alpha \longleftrightarrow \text{entropy / exploration}$$

$$\lambda \longleftrightarrow \text{constraint pressure}$$

With automatic entropy tuning,

$$\mathcal{H}(\pi) - \mathcal{H}_{\text{target}} \longrightarrow \alpha$$

while the Lagrangian update responds to

$$J_C - d \longrightarrow \lambda.$$

Entropy–Reward Coverage

Measure the regions of the policy's entropy–value space that have been exercised:

$$\mathcal{C}_{ER} = \mathcal{C}(\mathcal{H}(\pi(\cdot|s)), Q_R(s, a))$$

Each state-action pair during testing will contribute a point in this 2-D space. We divide the space into bins and check how many regions have been exercised by the test suite.

Why do we need it?

Standard state/action coverage does not distinguish between different exploration–exploitation regimes.

Entropy	Value	Interpretation
High	High	Exploration among promising actions
High	Low	Exploratory but potentially inefficient
Low	High	Confident exploitation
Low	Low	Potentially problematic policy behavior

Testing objective:

Ensure that testing exercises the important combinations of policy stochasticity and action value.

Reward–Constraint Coverage

Definition

Measure coverage of the reward–cost decision space:

$$\mathcal{C}_{RC} = \mathcal{C}(Q_R(s, a), Q_C(s, a))$$

The policy encounters qualitatively different decision regimes:

Reward	Cost	Interpretation
High	Low	Desirable action
Low	Low	Safe but unattractive
High	High	Critical reward–safety tradeoff
Low	High	Undesirable action

The critical region is:

$$Q_R(s, a) \text{ high} \quad \text{and} \quad Q_C(s, a) \text{ high}$$

because the policy must trade reward against constraint satisfaction.

Why do we need it?

A conventional coverage criterion may show that an action/state was visited, but not whether the policy was tested in situations where **high reward conflicts with safety**.

A New Safety-Coverage Hierarchy

Motivation: Conventional DRL coverage focuses on raw state/action visitation. For constrained SAC, testing must prioritize **safety-relevant regions**.

State $\longrightarrow$ Action $\longrightarrow$ Policy



Safety State $\longrightarrow$ Safety-Action $\longrightarrow$ Safety-Policy

Safety State Coverage

Partition states by proximity to constraint violations:

Safe $\rightarrow$ Critical
 $\rightarrow$ Unsafe

Safety-Action Coverage

Partition actions by their safety consequences:

$a_{\text{safe}} \rightarrow a_{\text{critical}}$
 $\rightarrow a_{\text{unsafe}}$

Safety-Policy Coverage

Evaluate policy distribution across safety regimes:

$\pi(a|s)$
and $P_{\text{unsafe}}(s)$

Key Idea

Coverage should measure how thoroughly the policy has been evaluated in safety-critical operational regimes.

Safety State Coverage: Margin & Partitioning

Motivation

- Standard state coverage tracks raw visitation without distinguishing risk.
- For constrained SAC, states must be qualified by their **safety criticality**.

Safety Cost-to-Go & Safety Margin

Let $V_C(s) = \mathbb{E}_{a \sim \pi}[Q_C(s, a)]$ denote expected cumulative cost. Given cost budget d :

$$M(s) = d - V_C(s)$$

Safety Margin Partitioning

Safety Margin	State Region	Interpretation
$M(s) \gg 0$	Safe	Nominal operating margin
$M(s) > 0$	Near-safe	Approaching constraint boundary
$M(s) \approx 0$	Boundary	Critical boundary region
$M(s) < 0$	Unsafe	Predicted constraint violation

Safety State Coverage: Metric & Objectives

Coverage Metric

$$C_{ss} = \frac{|\{\text{covered safety-state regions}\}|}{|\{\text{defined safety-state regions}\}|}$$

Primary Testing Objective

Testing must not merely evaluate *where the agent can go* under nominal conditions, but specifically expose states that are near or beyond the safety boundary.

Safety-Action Coverage: Partitioning

Motivation

- Continuous actions must be grouped by their **safety consequences**.

Action Space Partitioning

Let $C(s, a)$ denote immediate cost (or safety surrogate $Q_C(s, a)$):

$$\mathcal{A}_{\text{safe}} = \{a : C(s, a) < \tau_{\text{safe}}\}$$

$$\mathcal{A}_{\text{critical}} = \{a : |C(s, a) - \tau_{\text{safe}}| \leq \epsilon\}$$

$$\mathcal{A}_{\text{unsafe}} = \{a : C(s, a) > \tau_{\text{safe}}\}$$

Core Principle

Two distinct continuous actions provide identical coverage if they belong to the same safety equivalence class.

Safety-Action Coverage: Metric & Objectives

Coverage Metric

$$C_{SA} = \frac{|\{\text{covered safety-action regions}\}|}{|\{\text{defined safety-action regions}\}|}$$

Testing Objectives by Region

Action Region	Safety Level	Testing Objective
$\mathcal{A}_{\text{safe}}$	Safe	Verify nominal performance
$\mathcal{A}_{\text{critical}}$	Boundary	Safety margin
$\mathcal{A}_{\text{unsafe}}$	Unsafe	Expose direct violations

SAC Coverage: Unsafe Action Probabilities

Motivation

- SAC deploys a stochastic policy: $a \sim \pi(\cdot|s)$.
- Evaluating a single sampled action fails to expose tail-risk failure modes.

Safety Formulations

$$P_{\text{unsafe}}(s) = \int_{\mathcal{A}_{\text{unsafe}}} \pi(a|s) da, \quad P_{\text{critical}}(s) = \int_{\mathcal{A}_{\text{critical}}} \pi(a|s) da$$

Policy-Level Risk Stratification

$P_{\text{unsafe}}(s)$	Policy Behavior	Interpretation
≈ 0	Almost always safe	Low risk
0.01 – 0.05	Rare unsafe actions	Potential risk
0.05 – 0.20	Frequent unsafe actions	High risk
> 0.20	Unsafe actions likely	Critical failure